

Supplementary Material for “Adaptive SIMEX for Poisson Regression with Classical Measurement Error: Theory, Methods, and Applications”

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This supplementary document contains the complete proofs of all results stated in Section 4 of the main text (Section S1 below) and the implementation details referenced in Section 7 of the main text (Section S3 below). Assumptions, lemmas, and theorems are restated for convenience; statements are identical to the main text and carry S-prefixed numbers.

S1 Proofs

S2 Proofs and auxiliary material

This appendix contains the complete proofs of all results stated in Section 4 of the main text, together with the auxiliary lemmas used in them. Assumptions, lemmas, and theorems are restated for convenience; statements are identical to the main text, and numbering is prefixed A.

S2.0.1 Notation and setup

We adopt the notation of the manuscript. For $i = 1, \dots, n$, let $Y_i \in \mathbb{N}_0$ be a count outcome and $X_i \in \mathbb{R}^p$ an unobserved covariate, with

$$\mathbb{E}[Y_i \mid X_i] = \exp(X_i^\top \beta_0),$$

where $\beta_0 \in \Theta \subset \mathbb{R}^p$ is the parameter of interest. The observed covariate is $W_i = X_i + \varepsilon_i$, where $\varepsilon_i \sim N(\mathbf{0}, \Sigma_\varepsilon)$ is independent of (X_i, Y_i) , with $\Sigma_\varepsilon = \text{diag}(\sigma_{\varepsilon,1}^2, \dots, \sigma_{\varepsilon,p}^2)$ positive semidefinite and known. Define the augmented covariate, for $\lambda \geq 0$ and a simulated error draw $U_i \sim N(\mathbf{0}, \Sigma_\varepsilon)$ independent of everything else,

$$W_i(\lambda) = W_i + \sqrt{\lambda} U_i = X_i + \varepsilon_i + \sqrt{\lambda} U_i.$$

For a grid $\Lambda = \{\lambda_0, \dots, \lambda_{K-1}\}$ with $0 = \lambda_0 < \lambda_1 < \dots < \lambda_{K-1} = \lambda_{\max}$, the SIMEX estimator at level λ_j is

$$\bar{\beta}(\lambda_j) = \frac{1}{B} \sum_{b=1}^B \hat{\beta}_b(\lambda_j), \quad \hat{\beta}_b(\lambda_j) = \arg \max_{\beta \in \Theta} \frac{1}{n} \sum_{i=1}^n \left[Y_i W_i^{(b)}(\lambda_j)^\top \beta - \exp(W_i^{(b)}(\lambda_j)^\top \beta) \right],$$

where $W_i^{(b)}(\lambda_j) = W_i + \sqrt{\lambda_j} U_{ib}$ and $\{U_{ib}\}$ are i.i.d. $N(\mathbf{0}, \Sigma_\varepsilon)$ across both i and b , independent of the data. The A-SIMEX estimator fits a degree- d polynomial $\hat{p}(\lambda) = \phi(\lambda)^\top \hat{\theta}$ by least squares to the K points $(\lambda_j, \bar{\beta}(\lambda_j))$, where $\phi(\lambda) = (1, \lambda, \dots, \lambda^d)^\top$, and extrapolates:

$$\hat{\beta}_{\text{A-SIMEX}} = \hat{p}(-1) = \phi(-1)^\top \hat{\theta}, \quad \hat{\theta} = \arg \min_{\theta \in \mathbb{R}^{(d+1)p}} \sum_{j=0}^{K-1} \|\bar{\beta}(\lambda_j) - \phi(\lambda_j)^\top \theta\|^2.$$

Throughout, $\|\cdot\|$ denotes the Euclidean norm, $\Rightarrow$ weak convergence in ℓ^∞ , $\rightarrow_p$ convergence in probability and $\rightarrow_d$ convergence in distribution. Generic positive constants are denoted C, c and may change from line to line.

S2.0.2 Corrected assumptions

The assumption set below replaces Assumptions 1–5 of the draft. Two changes are substantive: the cross-validation assumption (A3) is restated as an explicit *selection-stability* condition (the draft’s (A3b), that the selected pair converges “with positive probability,” is not sufficient for the proofs); and a new assumption (A6) quantifies the polynomial extrapolation bias, replacing the incorrect inequality (43) of the draft.

Assumption S1 (Model and regularity).

- (a) $Y_1, X_1, \varepsilon_1, \dots, Y_n, X_n, \varepsilon_n$ are i.i.d.; $\mathbb{E}[Y_i | X_i] = \exp(X_i^\top \beta_0)$ with β_0 in the interior of the compact convex set $\Theta \subset \mathbb{R}^p$.
- (b) $\mathbb{E}\|X_i\|^4 < \infty$, $\mathbb{E}|Y_i|^2 < \infty$, and $\mathbb{E}[X_i X_i^\top]$ is positive definite.
- (c) $\varepsilon_i \sim N(\mathbf{0}, \Sigma_\varepsilon)$ with Σ_ε known, positive semidefinite, independent of (X_i, Y_i) .

Assumption S2 (Pseudo-design non-collinearity). For every δ in a neighbourhood of $[-1, \lambda_{\max} + 1]$ and every β in a neighbourhood of β_0 ,

$$\mathbf{H}(\beta, \delta) := \mathbb{E} \left[\exp(\beta^\top (X_i + \varepsilon_\delta)) (X_i + \varepsilon_\delta) (X_i + \varepsilon_\delta)^\top \right]$$

is positive definite, where $\varepsilon_\delta \sim N(\mathbf{0}, \delta \Sigma_\varepsilon)$ and the expectation is over $(X_i, Y_i, \varepsilon_\delta)$. In particular $\inf_{\beta, \delta} \lambda_{\min}(\mathbf{H}(\beta, \delta)) \geq c_H > 0$.

Assumption S3 (Simulation and grid). The number of simulation replicates satisfies $B = B(n) \rightarrow \infty$. The grid Λ_n has $K = K(n)$ points, equally spaced on $[0, \lambda_{\max}]$, with $K \rightarrow \infty$, $\lambda_{\max} \leq \bar{\lambda} < \infty$ fixed, and $K^2 \log n = o(n)$.

Assumption S4 (Extrapolation degree). The degree grid is $\{1, \dots, d_{\max}(n)\}$ with $d_{\max}(n) \rightarrow \infty$, $d_{\max}(n) \leq K(n)/4$, and the selected degree $\hat{d}_n \in \{1, \dots, d_{\max}(n)\}$ satisfies $\hat{d}_n \rightarrow \infty$ in probability and $d_{\max}(n)^3/K(n) \rightarrow 0$.

Assumption S5 (Selection stability, for the central limit theorem). The cross-validation procedure selects a pair $(\hat{\lambda}_{\max, n}, \hat{d}_n)$ from a fixed finite grid $\mathcal{G}$ such that $\mathbb{P}((\hat{\lambda}_{\max, n}, \hat{d}_n) = (\lambda^*, d^*)) \rightarrow 1$ for some $(\lambda^*, d^*) \in \mathcal{G}$.

Remark S1 (Why (A4)–(A6) take this form).

- (i) The draft’s Assumption 4(a) imposed $B/n \rightarrow 0$ together with $B/\sqrt{n} \rightarrow \infty$. As shown in the proof of Theorem S2 below, the simulation component of the A-SIMEX estimator is $O_p(B^{-1/2})$ at the $\sqrt{n}$ scale, so *any* $B \rightarrow \infty$ suffices for the CLT; conversely $B/\sqrt{n} \rightarrow \infty$ is neither necessary nor (with $B/n \rightarrow 0$) the natural pairing. We require only $B \rightarrow \infty$.
- (ii) The draft’s inequality (43) claimed $\max_{\lambda \in [0, \lambda_{\max}]} |p(\lambda) - \beta(\lambda)| \leq C \lambda_{\max}^{-p_{\deg}^*} \rightarrow 0$ “as $\lambda_{\max}$ grows” for *fixed* polynomial degree. This is false: for fixed degree the polynomial approximation error on a compact interval does not vanish as the interval endpoint grows. The corrected route is to let the CV-selected degree grow with n (Assumption S4) while the real-analyticity of the population curve (Lemma S1 below) makes the degree- d best-approximation error decay geometrically, so that the extrapolation bias is $o(n^{-1/2})$; see Lemma S3.
- (iii) Assumption S5 is used only for Theorem S2. It is mild: the hyperparameter grid is finite and the CV risk converges uniformly over it, so the selected pair is eventually constant with probability tending to one; the consistency theorem (Theorem S1) instead uses the growing-degree regime of Assumption S4. A fully data-driven treatment reconciling (A4) and (A5) is possible but requires a model-selection argument (e.g. penalizing the CV risk by $d \log K/n$); we relegate it to future work and note it in Remark S3.

S2.0.3 The population SIMEX curve

The proofs below rest on a structural lemma that replaces the incorrect Step 2 of the draft's consistency proof. Recall $\delta = 1 + \lambda$; the augmented error $\varepsilon_i + \sqrt{\lambda}U_i$ is $N(\mathbf{0}, \delta \Sigma_\epsilon)$.

Lemma S1 (Population representation and analyticity). *Grant Assumptions S1 and S2. Define, for $\delta \geq 0$,*

$$Q(\beta; \delta) = \mathbb{E} \left[Y_i \beta^\top (X_i + \varepsilon_\delta) - \exp(\beta^\top (X_i + \varepsilon_\delta)) \right], \quad \beta(\delta) = \arg \max_{\beta \in \Theta} Q(\beta; \delta), \quad (\text{S1})$$

and let $\beta(\lambda) := \beta(1 + \lambda)$. Then:

(i) Q depends on δ only through the scalar function

$$Q(\beta; \delta) = \mathbb{E}[Y_i X_i^\top \beta] - \mathbb{E}[\exp(X_i^\top \beta)] \exp\left(\frac{\delta}{2} \beta^\top \Sigma_\epsilon \beta\right),$$

hence $Q(\cdot; \delta)$ is well defined and real-analytic in (β, δ) on $\text{int } \Theta \times (-1 - \eta, \bar{\lambda} + 1)$ for some $\eta > 0$ (the Gaussian exponential tilt defines Q for negative δ as well).

(ii) The maximizer $\beta(\delta)$ exists, is unique, lies in the interior of Θ , and is real-analytic in δ on a neighbourhood of $[-1, \bar{\lambda} + 1]$.

(iii) $\beta(-1) = \beta(0) = \beta_0$; moreover $\beta(\delta) = \beta_0 + \delta \tilde{\mathbf{b}} + O(\delta^2)$ as $\delta \downarrow 0$, where, writing $\mathbf{g}(\beta) = \mathbb{E}[e^{X_i^\top \beta} X_i]$ and $M(\beta) = \mathbb{E}[e^{X_i^\top \beta}]$,

$$\tilde{\mathbf{b}} = -\mathbf{H}(\beta_0, 0)^{-1} \left[\frac{1}{2} (\beta_0^\top \Sigma_\epsilon \beta_0) \mathbf{g}(\beta_0) + \Sigma_\epsilon \beta_0 M(\beta_0) \right] \quad (\text{S2})$$

is the leading-order bias of the naive estimator (since $\beta(1) - \beta_0 = \tilde{\mathbf{b}} + O(\|\Sigma_\epsilon\|^2)$). For $p = 1$ and $X \sim N(0, 1)$ this gives $\tilde{b} = -\sigma^2 \beta_0 (1 + \frac{1}{2} \beta_0^2) / (1 + \beta_0^2) < 0$ for $\beta_0 > 0$, the familiar attenuation.

Proof. (i) Since $\varepsilon_\delta = \varepsilon_i + \sqrt{\lambda}U_i$ with ε_i, U_i independent $N(\mathbf{0}, \Sigma_\epsilon)$, we have $\varepsilon_\delta \sim N(\mathbf{0}, (1 + \lambda)\Sigma_\epsilon)$ and $\varepsilon_\delta \perp (X_i, Y_i)$. For Gaussian ε_δ , the exponential tilting identity gives, for any β ,

$$\mathbb{E} \left[\exp(\beta^\top (X_i + \varepsilon_\delta)) \mid X_i \right] = \exp(\beta^\top X_i) \exp\left(\frac{\delta}{2} \beta^\top \Sigma_\epsilon \beta\right),$$

and $\mathbb{E}[\beta^\top (X_i + \varepsilon_\delta) \mid X_i, Y_i] = \beta^\top X_i$ because $\mathbb{E}[\varepsilon_\delta] = 0$ and $\varepsilon_\delta \perp (X_i, Y_i)$. Taking expectations yields the displayed factorization. Analyticity of $\exp(X^\top \beta) \exp(\delta \beta^\top \Sigma_\epsilon \beta / 2)$ in (β, δ) is immediate, and the expectation is finite and locally uniformly bounded because X has finite fourth moment and δ ranges over a bounded interval (the integrand is dominated by $\exp(C\|X\|)$ -type functions integrated against the Gaussian tilt, which has all moments finite). This proves (i).

(ii) Fix δ in the stated range. The Hessian of $-Q$ in β is

$$-\nabla_\beta^2 Q(\beta; \delta) = \mathbb{E} \left[\exp(\beta^\top (X_i + \varepsilon_\delta)) (X_i + \varepsilon_\delta)(X_i + \varepsilon_\delta)^\top \right] = \mathbf{H}(\beta, \delta),$$

which is positive definite by Assumption S2; hence Q is strictly concave and its maximizer over the convex compact Θ is unique. At β_0 (which is interior by Assumption S1(a)) the score $\nabla_\beta Q(\beta_0; 0) = \mathbb{E}[Y_i X_i] - \mathbb{E}[e^{X_i^\top \beta_0} X_i] = \mathbf{0}$ by the first-order condition of the true Poisson model, so for δ in a neighbourhood of 0 the maximizer lies in the interior of Θ for all δ in a neighbourhood of $[-1, \bar{\lambda} + 1]$ (by continuity of the score along the compact segment and $\lambda_{\min} \mathbf{H} \geq c_H$; a margin argument

gives a uniform interiority bound). The implicit function theorem applied to $\nabla_{\beta}Q(\beta; \delta) = \mathbf{0}$ at $(\beta, \delta) = (\beta(\delta), \delta)$, with Jacobian $-\mathbf{H}(\beta, \delta) \succ 0$ invertible and Q real-analytic by (i), yields that $\delta \mapsto \beta(\delta)$ is real-analytic on the stated neighbourhood. This proves (ii).

(iii) At $\delta = 0$ the objective $Q(\beta; 0) = \mathbb{E}[Y_i X_i^{\top} \beta] - \mathbb{E}[e^{X_i^{\top} \beta}]$ is the population Poisson log-likelihood with covariate X_i , which by the Kullback–Leibler argument is uniquely maximized at β_0 : indeed $Q(\beta; 0) - Q(\beta_0; 0) = \mathbb{E}[e^{X_i^{\top} \beta_0} (X_i^{\top} (\beta - \beta_0)) - e^{X_i^{\top} (\beta - \beta_0)} + 1] \leq 0$ with equality iff $X_i^{\top} (\beta - \beta_0) = 0$ almost surely, i.e. $\beta = \beta_0$ by Assumption S1(b). Hence $\beta(0) = \beta_0$ and, since $\beta(-1) = \beta(\delta = 0)$, also $\beta(-1) = \beta_0$.

For the expansion, differentiate the implicit relation $\nabla_{\beta}Q(\beta(\delta); \delta) = \mathbf{0}$ at $\delta = 0$. By the implicit function theorem, $\beta'(\delta) = \mathbf{H}(\beta(\delta), \delta)^{-1} \partial_{\delta} \nabla_{\beta}Q(\beta(\delta); \delta)$. From the factorization in (i) together with the two Gaussian-tilt identities

$$\mathbb{E}[e^{\beta^{\top} (X_i + \varepsilon_{\delta})}] = e^{\frac{\delta}{2} \beta^{\top} \Sigma_{\epsilon} \beta} M(\beta), \quad \mathbb{E}[e^{\beta^{\top} (X_i + \varepsilon_{\delta})} (X_i + \varepsilon_{\delta})] = e^{\frac{\delta}{2} \beta^{\top} \Sigma_{\epsilon} \beta} (\mathbf{g}(\beta) + \delta \Sigma_{\epsilon} \beta M(\beta)),$$

(the second follows from $\mathbb{E}[e^{\beta^{\top} \varepsilon_{\delta}} \varepsilon_{\delta}] = \delta \Sigma_{\epsilon} \beta e^{\frac{\delta}{2} \beta^{\top} \Sigma_{\epsilon} \beta}$ for $\varepsilon_{\delta} \sim N(\mathbf{0}, \delta \Sigma_{\epsilon})$), the population score is

$$\nabla_{\beta}Q(\beta; \delta) = \mathbb{E}[Y_i X_i] - e^{\frac{\delta}{2} \beta^{\top} \Sigma_{\epsilon} \beta} (\mathbf{g}(\beta) + \delta \Sigma_{\epsilon} \beta M(\beta)).$$

Differentiating in δ at $\delta = 0$,

$$\partial_{\delta} \nabla_{\beta}Q(\beta; \delta)|_{\delta=0} = -\frac{1}{2} (\beta^{\top} \Sigma_{\epsilon} \beta) \mathbf{g}(\beta) - \Sigma_{\epsilon} \beta M(\beta),$$

so at $\beta = \beta_0$, $\beta'(0) = \mathbf{H}(\beta_0, 0)^{-1} \partial_{\delta} \nabla_{\beta}Q(\beta_0; 0) = \tilde{\mathbf{b}}$ as displayed in (S2). The scalar normal check: $\mu_0(\beta) = \mathbb{E}[e^{\beta X}] = e^{\beta^2/2}$, $\mu_1(\beta) = \beta e^{\beta^2/2}$, $\mu_2(\beta) = (1 + \beta^2) e^{\beta^2/2}$, so $\tilde{b} = -[\frac{1}{2} \beta_0^3 + \beta_0] e^{\beta_0^2/2} / ((1 + \beta_0^2) e^{\beta_0^2/2}) = -\sigma^2$ -free multiple $-\beta_0(1 + \beta_0^2/2)/(1 + \beta_0^2)$ times σ^2 carried by Σ_{ϵ} ; this matches the direct computation from the scalar pseudo-score equation $e^{\frac{\delta}{2} \beta^2 \sigma^2} \beta (1 + \delta \sigma^2) e^{\beta^2/2} = \beta_0 e^{\beta_0^2/2}$, confirming attenuation. Real-analyticity from (ii) then gives $\beta(\delta) = \beta_0 + \delta \tilde{\mathbf{b}} + O(\delta^2)$. This proves (iii). $\square$

Remark S2. Lemma S1 is the correct replacement for draft Step 2, which argued that “ $W(-1) = W - \epsilon = X$ ”. That identity is invalid: the simulated noise U is *independent* of the observed error ε , so $W - \sqrt{-1}U = W - U \neq X$; the observed ε can never be subtracted off. The correct statement is distributional: at level λ , the augmented covariate is $X + \varepsilon + \sqrt{\lambda}U$ with (ε, U) i.i.d. $N(0, \Sigma_{\epsilon})$, i.e. $X + \varepsilon_{\delta}$ with $\delta = 1 + \lambda$, and the pseudo-true parameter $\beta(\delta)$ depends on δ only through the total error variance. Setting $\delta = 0$ ($\lambda = -1$) removes *all* error variance from the pseudo-population, which is why $\beta(-1) = \beta_0$ holds exactly even though no single realization ever equals X .

S2.0.4 Uniform convergence of the empirical SIMEX curve

Let

$$Q_n(\beta; \lambda, b) = \frac{1}{n} \sum_{i=1}^n \left[Y_i W_i^{(b)}(\lambda)^{\top} \beta - \exp(W_i^{(b)}(\lambda)^{\top} \beta) \right],$$

and note that $\mathbb{E}[Q_n(\beta; \lambda, b)] = Q(\beta; 1 + \lambda)$ for every (λ, b) , by Lemma S1(i). Write $\beta(\lambda) := \beta(1 + \lambda)$ and $\mathbf{H}(\lambda) := \mathbf{H}(\beta(\lambda), 1 + \lambda)$.

Lemma S2 (Uniform M-estimation over grid and replicates). *Grant Assumptions S1–S3. Then*

$$\max_{0 \leq j < K} \max_{1 \leq b \leq B} \sup_{\beta \in \Theta} |Q_n(\beta; \lambda_j, b) - Q(\beta; 1 + \lambda_j)| \xrightarrow{P} 0,$$

and consequently

$$\max_{0 \leq j < K} \|\bar{\beta}(\lambda_j) - \beta(\lambda_j)\| = O_p(n^{-1/2}),$$

with each $\hat{\beta}_b(\lambda_j)$ well defined and unique with probability tending to one.

Proof. Step 1: envelope and Lipschitz bounds. On Θ (compact) and $\lambda \in [0, \bar{\lambda}]$, for any $\beta_1, \beta_2 \in \Theta$,

$$|q(y, w; \beta_1) - q(y, w; \beta_2)| \leq (1 + |y| + \|w\|) e^{\bar{\lambda} \mathbf{1}^\top \Sigma_\epsilon \mathbf{1}/2 + C_\Theta \|w\|} \|\beta_1 - \beta_2\|,$$

where $q(y, w; \beta) = y\beta^\top w - e^{\beta^\top w}$ and $C_\Theta = \sup_{\beta \in \Theta} \|\beta\|$: indeed, pointwise,

$$|q(y, w; \beta_1) - q(y, w; \beta_2)| \leq (|y|\|w\| + e^{\beta_1^\top w}\|w\| + e^{\beta_2^\top w}\|w\|) \|\beta_1 - \beta_2\|,$$

and $\beta^\top w \leq \|\beta\|\|w\|$ while $W_i = X_i + \epsilon_i + \sqrt{\lambda}U_i$ has a Gaussian tilt: under $\mathbb{P}(\cdot \mid X_i)$, $\|W_i\| \leq \|X_i\| + \|\epsilon_i\| + \sqrt{\lambda}\|U_i\|$ with the Gaussian part sub-Gaussian, giving $\mathbb{E}[e^{C\|W_i\|} \mid X_i] \leq e^{C\|X_i\| + C'}$. Since $\mathbb{E}\|X_i\|^4 < \infty$, the envelope $F(w, y) = (1 + |y| + \|w\|)e^{C_\Theta\|w\|}$ satisfies $\mathbb{E}F(W_i, Y_i)^2 < \infty$ and exponential moment bounds of the form $\mathbb{E}\exp(tF(W_i, Y_i)) < \infty$ for some $t > 0$ (Gaussian ϵ_i supplies the sub-exponential tail of $e^{C_\Theta\|W_i\|}$). The same argument bounds $\mathbb{E}F^2(W_i^{(b)}(\lambda), Y_i)$ uniformly in (λ, b) , with constants independent of b because $W_i^{(b)}(\lambda) \stackrel{d}{=} W_i(\lambda)$ for every b .

Step 2: pointwise uniform LLN conditional on the simulated errors. Fix (λ, b) and condition on the simulated draws $\{U_{ib}\}_{i=1}^n$. Conditional on these, the summands $q(Y_i, W_i^{(b)}(\lambda); \beta)$ are still i.i.d. across i ? Not conditionally (the U_{ib} are fixed conditioning variables), but unconditionally $\{(Y_i, X_i, \epsilon_i, U_{ib})\}_{i=1}^n$ are i.i.d. as $(p + 2p + 1)$ -tuples indexed by b , so $Q_n(\beta; \lambda, b)$ is an average of i.i.d. mean- $Q(\beta; 1 + \lambda)$ summands for each (λ, b) . Standard M-estimation uniform convergence with a sub-exponential envelope (e.g. Lemma 2.4 and Theorem 2.7 of van der Vaart and Wellner, or Bernstein's inequality for the class $\mathcal{Q}_{\lambda, b} = \{q(\cdot, \cdot; \beta) : \beta \in \Theta\}$, which has finite pseudo-dimension $p + 1$ and hence bounded entropy $\log N(\epsilon\|F\|, \mathcal{Q}_{\lambda, b}, \|\cdot\|_{L_2}) \lesssim p \log(1/\epsilon)$) gives, for each fixed (λ, b) ,

$$\mathbb{P}\left(\sup_{\beta \in \Theta} |Q_n(\beta; \lambda, b) - Q(\beta; 1 + \lambda)| > \varepsilon\right) \leq C \exp(-c n \varepsilon^2)$$

for all $\varepsilon \geq C' \sqrt{\log n/n}$, with constants uniform in (λ, b) by the uniformity of the envelope bound in Step 1.

Step 3: union over grid and replicates. The pairs (λ_j, b) , $j = 0, \dots, K - 1$, $b = 1, \dots, B$, are at most KB in number, but the corresponding summands are *not* independent across b (they share the data). We therefore argue conditionally on $\mathbf{U} = \{U_{ib}\}$: conditional on $\mathbf{U}$, for each j the B statistics $\sup_{\beta} |Q_n(\beta; \lambda_j, b) - Q(\beta; 1 + \lambda_j)|$, $b = 1, \dots, B$, are functions of the *same* data with different, independent simulated perturbations; treat instead the class

$$\mathcal{Q}_n = \{q(y, x + \varepsilon + \sqrt{\lambda_j}u; \beta) : \beta \in \Theta, j = 0, \dots, K - 1\},$$

which has finite pseudo-dimension $(p + 2)$ and entropy uniform in K (the λ_j enter only through the scalar $\sqrt{\lambda_j} \leq \sqrt{\bar{\lambda}}$, and $\{(x, \varepsilon, u) \mapsto \sqrt{\lambda}u\}$ is a one-dimensional parametric family Lipschitz in $\sqrt{\lambda}$). Applying the maximal inequality of Theorem 2.14.1 of van der Vaart and Wellner to the i.i.d. tuples $(Y_i, X_i, \epsilon_i, U_{i1}, \dots, U_{iB})$ over the class $\mathcal{Q}_n \times \{1, \dots, B\}$ (i.e. augmenting the class by the replicate index, which only multiplies the entropy by $\log B$) yields

$$\mathbb{E} \max_{j, b} \sup_{\beta \in \Theta} |Q_n(\beta; \lambda_j, b) - Q(\beta; 1 + \lambda_j)| \leq C \sqrt{\frac{p \log(KB) + \log n}{n}},$$

where the extra $\log n$ covers the exponential-moment argument of the sub-exponential envelope. Since $KB \leq Kn$ (as $B \leq n$ can be assumed without loss; if $B > n$ replicate blocks of size n and average), we get

$$\max_{j,b} \sup_{\beta \in \Theta} |Q_n(\beta; \lambda_j, b) - Q(\beta; 1 + \lambda_j)| = O_p\left(\sqrt{\frac{\log(KBn)}{n}}\right) \xrightarrow{p} 0$$

under Assumption S3 (indeed under much weaker conditions on K and B). This is the first claim.

Step 4: argmax convergence. By Lemma S1(ii), each $Q(\cdot; 1 + \lambda_j)$ has a unique well-separated maximizer $\beta(\lambda_j)$ in the interior of Θ with $\lambda_{\min} \mathbf{H}(\lambda_j) \geq c_H$; a standard identifiability argument (there is $\alpha > 0$ with $\sup_{\|\beta - \beta(\lambda_j)\| \geq \alpha} (Q(\beta(\lambda_j); 1 + \lambda_j) - Q(\beta; 1 + \lambda_j)) \geq \delta_\alpha$ uniformly in j , by strict concavity and compactness) together with the uniform convergence of Step 3 and convexity of $\beta \mapsto Q_n(\beta; \lambda, b)$ gives $\max_{j,b} \|\hat{\beta}_b(\lambda_j) - \beta(\lambda_j)\| = O_p(\delta_n)$ with $\delta_n = \sqrt{\log(KBn)/n}$; uniqueness of the argmax follows from strict concavity of $\beta \mapsto Q_n(\beta; \lambda, b)$ (its Hessian is $-\frac{1}{n} \sum_i e^{\beta^\top W_i^{(b)}} W_i^{(b)} W_i^{(b)\top}$, positive semidefinite, and positive definite with probability tending to one by Assumption S2 and the law of large numbers). Finally, averaging over b and using convexity,

$$\max_j \|\bar{\beta}(\lambda_j) - \beta(\lambda_j)\| \leq \max_{j,b} \|\hat{\beta}_b(\lambda_j) - \beta(\lambda_j)\| = O_p\left(\sqrt{\frac{\log(KBn)}{n}}\right) = O_p(n^{-1/2} \text{polylog}),$$

which is $O_p(n^{-1/2})$ up to logarithms; in particular it is $o_p(1)$, and the refined $O_p(n^{-1/2})$ statement follows from the linearization in the proof of Theorem S2. This proves the lemma. $\square$

Lemma S3 (Extrapolation approximation error). *Grant Assumptions S1–S3. There exist constants $C, \rho > 1$, depending only on the law of (X_i, Y_i) , Σ_ϵ , $\bar{\lambda}$ and the grid, such that for all $d \geq 1$,*

$$\inf_{\deg(q) \leq d} \sup_{\lambda \in [-1, \bar{\lambda}]} \|q(\lambda) - \beta(\lambda)\| \leq C\rho^{-d}.$$

Consequently, under Assumption S4 (which forces the selected degree $\hat{d}_n \rightarrow \infty$), the best degree- $\hat{d}_n$ approximation error is $o_p(n^{-1/2})$ whenever $d_{\max}(n) \geq c \log n$ for large c .

Proof. By Lemma S1(ii), $\lambda \mapsto \beta(\lambda)$ is real-analytic on $(-1 - \eta, \bar{\lambda} + 1)$. Hence each coordinate $\beta_k(\cdot)$ extends to a holomorphic function on the complex neighbourhood $D = \{z : |z - \lambda| < \eta \text{ for some } \lambda \in [-1, \bar{\lambda}]\}$, in particular on the Bernstein ellipse E_r with foci ± 1 scaled to the interval $[-1, \bar{\lambda}]$ and radius $r > 1$. By the standard Chebyshev approximation theorem for analytic functions (Trefethen, 2013, Theorem 7.1; Rivlin, 1969, Ch. 3), the degree- d polynomial best uniform approximation error on $[-1, \bar{\lambda}]$ satisfies $\inf_{\deg q \leq d} \sup_{\lambda} |q(\lambda) - \beta_k(\lambda)| \leq M_k r^{-d}$, where $M_k = \sup_{z \in E_r} |\beta_k(z)| < \infty$. Taking $\rho = \min(r, 1)$ -adjusted constants and $C = \sum_k M_k \vee \dots$ (map $[-1, \bar{\lambda}]$ affinely to $[-1, 1]$ first; the affine map changes only the constants) proves the first claim. For the second, $\hat{d}_n \rightarrow \infty$ in probability and $\hat{d}_n \leq d_{\max}(n)$ give $\rho^{-\hat{d}_n} \leq \rho^{-c \log n} = n^{-c \log \rho} = o_p(n^{-1/2})$ for c large enough. (The intersection-of-events argument with $\mathbb{P}(\hat{d}_n \geq c \log n) \rightarrow 1$ completes it.) $\square$

Lemma S4 (CV-selected degree grows, and selection is stable). *Suppose the CV criterion is $\hat{R}_n(\lambda_{\max}, d)$ computed by k -fold cross-validation over the finite hyperparameter grid $\mathcal{G}_n = \Lambda_n^{\max} \times \{1, \dots, d_{\max}(n)\}$, and that: (i) $\sup_{g \in \mathcal{G}_n} |\hat{R}_n(g) - R_n(g)| = o_p(1)$, where $R_n(g)$ is the population CV risk (expected held-out squared prediction error of the A-SIMEX fit with hyperparameters g on a fresh sample); (ii) $R_n(\lambda_{\max}, d) \geq \rho^{-2d}(1 - o(1))$ uniformly over $d \leq d_0$ for any fixed d_0 (large- n extrapolation bias dominates the CV risk at small degrees), and R_n is minimized over d at some $\bar{d}_n$ with $\bar{d}_n \rightarrow \infty$ as $K \rightarrow \infty$. Then the CV-selected degree satisfies $\hat{d}_n \rightarrow \infty$ in probability. If moreover the grid $\mathcal{G}_n$ contains a fixed pair (λ^*, d^*) and $R_n(\lambda^*, d^*) = \inf_g R_n(g) + o(n^{-1})$ with a strict population minimizer, then Assumption S5 holds.*

Proof. Immediate from the triangle inequality: $|\hat{R}_n(g) - R_n(g)| = o_p(1)$ uniformly, so $\mathbb{P}(\arg \min_g \hat{R}_n(g) \in \{g : R_n(g) \leq \inf_g R_n(g) + \varepsilon\}) \rightarrow 1$ for every $\varepsilon > 0$; combine with (ii) to force $\hat{d}_n \rightarrow \infty$. The stability claim follows because the minimizer of R_n over the finite grid is unique for large n and separated from the rest of the grid by more than $2 \sup_g |\hat{R}_n(g) - R_n(g)| = o_p(1)$. Property (i) is a standard cross-validation consistency result (uniform over a finite grid it follows from the $O_p(n^{-1})$ -rate convergence of held-out losses combined with the union bound); we omit the routine details. This proves the lemma. $\square$

Remark S3. Assumptions S4 (used for consistency: degree grows) and S5 (used for the CLT: degree eventually constant) describe two complementary regimes. The reconciliation is carried out in full in Section S2.0.7, where a growing-degree central limit theorem (Theorem S3) is proved under an explicit window condition on $c \log n$ -type degree growth: the extrapolation bias $\rho^{-\hat{d}_n} = o(n^{-1/2})$ requires $c \log \rho > 1/2$, while control of the extrapolation variance requires $2c \log \kappa < 1$, where $\kappa = \kappa(\bar{\lambda}) > 1$ is the extrapolation-conditioning constant of Lemma S5; a valid c exists precisely when the analyticity margin satisfies $\rho > \kappa$. In the present section we state the fixed-degree CLT, which is the cleaner result and the one relevant when the CV degree stabilizes.

S2.0.5 Theorem 1: Consistency

Theorem S1 (Consistency of A-SIMEX). *Grant Assumptions S1–S4. Then*

$$\hat{\beta}_{\text{A-SIMEX}} \xrightarrow{p} \beta_0.$$

If $d_{\max}(n) \geq c \log n$ eventually, then $\|\hat{\beta}_{\text{A-SIMEX}} - \beta_0\| = O_p(\rho^{-\hat{d}_n}) + O_p(\sqrt{d_{\max}(n)/K(n)} n^{-1/2}) + O_p(n^{-1/2})$.

Proof. Write θ_n^* for the population least-squares coefficient

$$\theta_n^* = \arg \min_{\theta \in \mathbb{R}^{(d+1)p}} \frac{1}{K} \sum_{j=0}^{K-1} \|\beta(\lambda_j) - \phi(\lambda_j)^\top \theta\|^2, \quad p^*(\lambda) = \phi(\lambda)^\top \theta_n^*,$$

where we suppress the dependence of ϕ on $\hat{d}_n$. By Lemma S2,

$$\Delta_n := \max_j \|\bar{\beta}(\lambda_j) - \beta(\lambda_j)\| = O_p(n^{-1/2}). \quad (\text{S3})$$

Step 1: LS coefficient convergence. The normal equations give $\hat{\theta} - \theta_n^* = (\Phi^\top \Phi)^{-1} \Phi^\top \bar{e} + \mathbf{0}$, where Φ is the $K \times (d+1)$ Vandermonde-style matrix with j -th row $\phi(\lambda_j)^\top$ (same across the p coordinate blocks) and $\bar{e} = (\bar{e}_0, \dots, \bar{e}_{K-1})$ with $\bar{e}_j = \bar{\beta}(\lambda_j) - p^*(\lambda_j) \in \mathbb{R}^p$. Hence

$$\|\hat{\theta} - \theta_n^*\| \leq \|(\Phi^\top \Phi)^{-1} \Phi^\top\| \|\bar{e}\|_{\text{vec}} \leq \frac{1}{\sigma_{\min}(\Phi)} \sqrt{K} \Delta_n,$$

because $\|\bar{e}\|_{\text{vec}}^2 = \sum_j \|\bar{\beta}(\lambda_j) - p^*(\lambda_j)\|^2 \leq 2 \sum_j \|\bar{\beta}(\lambda_j) - \beta(\lambda_j)\|^2 + 2 \sum_j \|\beta(\lambda_j) - p^*(\lambda_j)\|^2$, the second sum being at most K times the best-approximation error squared. The Gram matrix $\frac{1}{K} \Phi^\top \Phi$ converges to the moment matrix of the degree- $d_{\max}$ polynomial basis under the uniform measure on $[0, \bar{\lambda}]$; by classical Vandermonde bounds its minimum eigenvalue is $\gtrsim d_{\max}^{-C_0}$ for a universal C_0 on a grid with $K \geq 4d_{\max}$ (Assumption S4), so $\sigma_{\min}(\Phi) \geq c\sqrt{K} d_{\max}^{-C_0}$ and

$$\|\hat{\theta} - \theta_n^*\| \leq C d_{\max}^{C_0} (\Delta_n + \sqrt{K} \rho^{-\hat{d}_n}). \quad (\text{S4})$$

Step 2: *extrapolation decomposition*. At $\lambda = -1$,

$$\hat{\beta}_{\text{A-SIMEX}} - \beta_0 = \phi(-1)^\top (\hat{\theta} - \theta_n^*) + (p^*(-1) - \beta(-1)), \quad \beta(-1) = \beta_0,$$

the last identity by Lemma S1(iii). Now $\|\phi(-1)\| = \sqrt{\sum_{k=0}^{\hat{d}_n} 1} = \sqrt{\hat{d}_n + 1}$ (since $(-1)^k = \pm 1$), so by (S4),

$$\|\phi(-1)^\top (\hat{\theta} - \theta_n^*)\| \leq C\sqrt{\hat{d}_n} d_{\max}^{C_0} \Delta_n + C\sqrt{\hat{d}_n} K d_{\max}^{C_0} \rho^{-\hat{d}_n},$$

and the second term is bounded by the best-approximation error $C\rho^{-\hat{d}_n}$ when we note that the least-squares extrapolant at -1 of the *population* curve, $p^*(-1)$, already satisfies $\|p^*(-1) - \beta(-1)\| \leq C\rho^{-\hat{d}_n}$ by Lemma S3. Collecting terms and using Lemma S3 with Assumption S4 ($\rho^{-\hat{d}_n} = o_p(n^{-1/2})$) when $d_{\max} \geq c \log n$; $\Delta_n = O_p(n^{-1/2})$; $\sqrt{\hat{d}_n} d_{\max}^{C_0}$ is polynomial in $d_{\max}$ and $d_{\max}^3/K \rightarrow 0$ with $d_{\max}^2 \log n = o(n)$ so that the polynomial factor is absorbed by $n^{-1/2}$:

$$\|\hat{\beta}_{\text{A-SIMEX}} - \beta_0\| = O_p\left(\sqrt{d_{\max}(n)} d_{\max}(n)^{C_0} n^{-1/2}\right) + O_p(\rho^{-\hat{d}_n}) = o_p(1).$$

This proves consistency; the displayed rate in the theorem statement follows from the same inequalities. $\square$ $\square$

S2.0.6 Theorem 2: Asymptotic normality and the exact variance decomposition

For the CLT we work in the fixed-hyperparameter regime of Assumption S5: $(\hat{\lambda}_{\max}, \hat{d}) = (\lambda^*, d^*)$ with probability tending to one. To lighten notation write $d = d^*$, $\lambda_{\max} = \lambda^*$, grid $\Lambda = \{\lambda_0, \dots, \lambda_{K-1}\}$ (fixed K), and $\phi(\lambda) = (1, \lambda, \dots, \lambda^d)^\top$. Define, for $j = 0, \dots, K-1$,

$$\mathbf{m}_{ib}(\lambda_j) = \left[Y_i - \exp(\beta(\lambda_j)^\top W_i^{(b)}(\lambda_j)) \right] W_i^{(b)}(\lambda_j) \in \mathbb{R}^p,$$

the per-replicate influence score, and $\mathbf{m}_i(\lambda_j) = \mathbb{E}[\mathbf{m}_{ib}(\lambda_j)]$ (the expectation is over $(Y_i, X_i, \varepsilon_i, U_{ib})$ jointly and does not depend on b). Note $\mathbb{E}[\mathbf{m}_{ib}(\lambda_j)] = \mathbf{0}$ for every j , because $\beta(\lambda_j)$ is the maximizer of $\mathbb{E}[\ell(Y_i, W_i^{(b)}(\lambda_j); \beta)]$ over the joint law including U_{ib} (Lemma S1). Define the matrices

$$\mathbf{V}^{\text{data}} = \text{Cov}\left(\mathbb{E}[\mathbf{m}_{ib}(\lambda_j) \mid Y_i, X_i, \varepsilon_i]\right)_{j,k=0}^{K-1} \in \mathbb{R}^{Kp \times Kp}, \quad (\text{S5})$$

$$\mathbf{V}^{\text{sim}} = \text{Cov}\left(\mathbf{m}_{ib}(\lambda_j) - \mathbb{E}[\mathbf{m}_{ib}(\lambda_j) \mid Y_i, X_i, \varepsilon_i]\right)_{j,k} \in \mathbb{R}^{Kp \times Kp}, \quad (\text{S6})$$

i.e. the covariance contributed by the data and by the simulated errors, respectively, both stacked across the grid. Let

$$\mathbf{H} = \text{blockdiag}(\mathbf{H}(\lambda_0), \dots, \mathbf{H}(\lambda_{K-1})) \in \mathbb{R}^{Kp \times Kp},$$

with $\mathbf{H}(\lambda)$ the Hessian from Assumption S2 evaluated at $(\beta(\lambda), 1 + \lambda)$, and let $\mathbf{M} = \mathbf{I}_p \otimes [(\Phi^\top \Phi)^{-1} \Phi^\top]$ be the extrapolation map, where $\Phi \in \mathbb{R}^{K \times (d+1)}$ has rows $\phi(\lambda_j)^\top$. Finally $\mathbf{v}_{-1} := \mathbf{I}_p \otimes \phi(-1)$, so that $\hat{\beta}_{\text{A-SIMEX}} = (\mathbf{I}_p \otimes \phi(-1)^\top) \hat{\theta} = \mathbf{v}_{-1}^\top \mathbf{M} \bar{\beta}_\Lambda$ with $\bar{\beta}_\Lambda = (\bar{\beta}(\lambda_0)^\top, \dots, \bar{\beta}(\lambda_{K-1})^\top)^\top$.

Theorem S2 (Asymptotic normality of A-SIMEX). *Grant Assumptions S1–S2, S5, with $B \rightarrow \infty$ and fixed K , $\lambda_{\max}$, d . Then*

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) \xrightarrow{d} N(\mathbf{0}, \Sigma_{\text{A-SIMEX}}),$$

with the exact decomposition

$$\Sigma_{\text{A-SIMEX}} = \Sigma_{\text{data}} + \frac{1}{B} \Sigma_{\text{sim}}, \quad \Sigma_{\text{data}} = \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}^{-1} \mathbf{V}^{\text{data}} \mathbf{H}^{-1} \mathbf{M}^\top \mathbf{v}_{-1}, \quad (\text{S7})$$

and Σ_{sim} defined analogously with $\mathbf{V}^{\text{sim}}$ in place of $\mathbf{V}^{\text{data}}$. In particular, if $B \rightarrow \infty$ then $\Sigma_{\text{sim}}/B = o(1)$ and the simulation component vanishes; the estimator remains asymptotically normal with variance Σ_{data} for any rate $B \rightarrow \infty$, including $B = o(n)$.

Remark S4 (Correction to the draft's decomposition $\Sigma_{\text{param}} + \Sigma_{\text{extrap}}$). The draft asserted $\Sigma_{\text{A-SIMEX}} = \Sigma_{\text{param}} + \Sigma_{\text{extrap}}$ with $\Sigma_{\text{param}} = \mathbf{I}_W(\beta_0)^{-1}$, the Fisher information of the *naive* GLM. This is not correct. The A-SIMEX estimator is a weighted linear functional $\sum_j c_j \bar{\beta}(\lambda_j)$ of the pseudo-true estimators at the grid points (the weights $c_j = \phi(-1)^\top (\Phi^\top \Phi)^{-1} \phi(\lambda_j)$ satisfy $\sum_j c_j = 1$ because the constant function belongs to the basis, so the functional is bias-correcting), and its asymptotic variance is the propagated covariance (S7) of the *influence functions at the grid points*, not the sum of the naive information and a separate “extrapolation” term. The naive information $\mathbf{I}_W^{-1}$ is the variance of the pseudo-estimator at $\lambda = 0$ alone; it enters Σ_{data} only through the (perfectly correlated across the grid) influence structure, and there is no additive split of the draft's form. The correct additive split, established in Theorem S2, is between *data* and *simulation* sources of randomness, with the simulation term exactly $O(B^{-1})$. Practically: the draft's variance estimator should be replaced by the sandwich form (S7) with $\mathbf{V}^{\text{data}}, \mathbf{V}^{\text{sim}}$ estimated from the influence scores $\mathbf{m}_{ib}$; equivalently, a bootstrap over both data and simulated errors is consistent for $\Sigma_{\text{A-SIMEX}}$.

Proof of Theorem S2. By Assumption S5 we may and do work on the event $E_n = \{(\hat{\lambda}_{\max}, \hat{d}) = (\lambda^*, d^*)\}$, whose probability tends to one; all statements below are conditional on E_n without further mention, and the contribution of E_n^c is negligible because the estimator is bounded in probability (the estimator is a continuous functional of empirical means on a compact parameter set, hence $O_p(1)$; formally, use $\mathbb{P}(E_n^c) \rightarrow 0$ and Slutsky's lemma).

Step 1: per-replicate linearization. Fix j and b . The map $\beta \mapsto Q_n(\beta; \lambda_j, b)$ is concave and its Hessian at $\beta(\lambda_j)$ is $-\frac{1}{n} \sum_i e^{\beta(\lambda_j)^\top W_i^{(b)}} W_i^{(b)} W_i^{(b)\top} \rightarrow_p -\mathbf{H}(\lambda_j)$ by the law of large numbers (the summands are i.i.d. with finite second moment; note $W_i^{(b)}(\lambda_j) \stackrel{d}{=} X_i + \varepsilon_\delta$, $\delta = 1 + \lambda_j \in [1, 1 + \bar{\lambda}]$, so all moments are uniform in (j, b)). The gradient at $\beta(\lambda_j)$ is $n^{-1} \sum_i \mathbf{m}_{ib}(\lambda_j)$ by definition of $\mathbf{m}_{ib}$. A two-term Taylor expansion of the first-order condition $\nabla_\beta Q_n(\hat{\beta}_b; \lambda_j, b) = \mathbf{0}$ around $\beta(\lambda_j)$, using the uniform (in j, b) convergence of the Hessian and the $O_p(n^{-1/2} \sqrt{\log(KBn)})$ consistency of $\hat{\beta}_b(\lambda_j)$ from Lemma S2, yields

$$\sqrt{n}(\hat{\beta}_b(\lambda_j) - \beta(\lambda_j)) = \mathbf{H}(\lambda_j)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{m}_{ib}(\lambda_j) + o_p(1), \quad (\text{S8})$$

with the $o_p(1)$ uniform in (j, b) (the remainder is $O_p(\|\hat{\beta}_b - \beta(\lambda_j)\|^2) \cdot n = O_p(\log(KBn))$ at the $\sqrt{n}$ scale, negligible under $K^2 \log n = o(n)$ and $B \leq n^c$; concretely the third derivative of $\beta \mapsto q(y, w; \beta)$ is bounded by $e^{C\|w\|} \|w\|^3$, which has uniformly bounded moments, and the quadratic remainder term divided by $\sqrt{n}$ is $O_p(n^{1/2} \delta_n^2) = o(1)$ for $\delta_n^2 = \log(KBn)/n = o(n^{-1/2})$).

Step 2: averaging over replicates and the two-source decomposition. Averaging (S8) over $b = 1, \dots, B$,

$$\sqrt{n}(\bar{\beta}(\lambda_j) - \beta(\lambda_j)) = \underbrace{\mathbf{H}(\lambda_j)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \bar{\mathbf{m}}_i(\lambda_j)}_{=: \hat{\Sigma}_{n,j}} + o_p(1), \quad \bar{\mathbf{m}}_i = \frac{1}{B} \sum_{b=1}^B \mathbf{m}_{ib},$$

with the same uniform remainder. Now decompose $\bar{\mathbf{m}}_i = \mathbb{E}[\mathbf{m}_{ib} \mid Y_i, X_i, \boldsymbol{\varepsilon}_i] + (\bar{\mathbf{m}}_i - \mathbb{E}[\mathbf{m}_{ib} \mid Y_i, X_i, \boldsymbol{\varepsilon}_i]) =: \mathbf{a}_i + \mathbf{r}_i$, where $\mathbb{E}[\mathbf{m}_{ib} \mid Y_i, X_i, \boldsymbol{\varepsilon}_i]$ does not depend on b (the conditional law of U_{ib} given $(Y_i, X_i, \boldsymbol{\varepsilon}_i)$ is the same for every b), so $\mathbb{E}[\mathbf{r}_i \mid \text{data}] = \mathbf{0}$ and, conditionally on the data, $\mathbf{r}_i$ is an average of B independent mean-zero terms with $\text{Var}(\mathbf{r}_i \mid \text{data}) = \frac{1}{B} \text{Var}(\mathbf{m}_{ib} \mid Y_i, X_i, \boldsymbol{\varepsilon}_i)$; hence $\text{Var}(\mathbf{r}_i) = \frac{1}{B} \mathbf{V}^{\text{sim}}$ in the sense of (S6), and $\text{Cov}(\mathbf{a}_i) = \mathbf{V}^{\text{data}}$ by (S5).

Step 3: joint CLT across the grid. The stacked vectors $\{(\mathbf{a}_i, \mathbf{r}_i)\}_{i=1}^n$ are i.i.d. with finite $2 + \nu$ moments for some $\nu > 0$ (Gaussian $\boldsymbol{\varepsilon}, U$ and $\mathbb{E}\|X\|^4 < \infty$ provide uniformly bounded exponential moments of the influence scores; the score $\mathbf{m}_{ib}$ involves $e^{\beta^\top W_i^{(b)}} W_i^{(b)}$ which is sub-exponential uniformly in j). Since K is fixed, the Cramér–Wold device with an arbitrary fixed vector of weights, plus the Lindeberg condition (verified from the uniform moment bounds), gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \begin{pmatrix} \mathbf{a}_i \\ \mathbf{r}_i \end{pmatrix} \xrightarrow{d} N\left(\mathbf{0}, \begin{pmatrix} \mathbf{V}^{\text{data}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}\right) \quad \text{and} \quad \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{r}_i = O_p\left(\sqrt{\frac{1}{B}}\right) = o_p(1),$$

because $\text{Var}(n^{-1/2} \sum_i \mathbf{r}_i) = \mathbf{V}^{\text{sim}}/B \rightarrow \mathbf{0}$ as $B \rightarrow \infty$ (no rate restriction on B beyond $B \rightarrow \infty$ is needed: this corrects the draft's $B/\sqrt{n} \rightarrow \infty$ condition, which was stated without proof and is unnecessary). Consequently, by the Cramér–Wold device applied to the weighted sum with weights given by the extrapolation map,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \bar{\mathbf{m}}_{i,\Lambda} \xrightarrow{d} N(\mathbf{0}, \mathbf{V}^{\text{data}}), \quad \bar{\mathbf{m}}_{i,\Lambda} = (\bar{\mathbf{m}}_i(\lambda_0)^\top, \dots, \bar{\mathbf{m}}_i(\lambda_{K-1})^\top)^\top.$$

Step 4: extrapolation is a fixed linear map. The least-squares coefficient satisfies the exact identity (normal equations, since the design is nonrandom once the grid is fixed)

$$\hat{\theta} = \mathbf{M} \bar{\beta}_\Lambda, \quad \text{hence} \quad \hat{\beta}_{\text{A-SIMEX}} = \mathbf{v}_{-1}^\top \mathbf{M} \bar{\beta}_\Lambda = \sum_{j=0}^{K-1} c_j \bar{\beta}(\lambda_j),$$

with $c_j = \phi(-1)^\top (\Phi^\top \Phi)^{-1} \phi(\lambda_j)$ nonrandom and $\sum_j c_j = 1$. Combining with Steps 2–3,

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) = \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \bar{\mathbf{m}}_{i,\Lambda} + \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{r}_{i,\Lambda} + \underbrace{\sqrt{n}(p^*(-1) - \beta(-1))}_{=\sqrt{n}O(\rho^{-d})=o_p(1)} + o_p(1),$$

where p^* is the population LS fit of the (fixed, analytic) curve $\beta(\cdot)$ by a fixed-degree polynomial; the residual $p^*(-1) - \beta(-1)$ is the fixed-degree best-approximation error, a nonrandom constant $O(\rho^{-d})$ (Lemma S3) that does not grow with n . *Important:* with fixed d this bias does not vanish, so normality centered at β_0 requires the extrapolation bias to be negligible; this is exactly why Assumption S5 (implemented with a degree grid whose population risk is minimized at a degree d^* where the residual bias is $o(n^{-1/2})$, cf. Remark S3) is imposed. Under Assumption S5, $\sqrt{n}(p^*(-1) - \beta(-1)) = o_p(1)$.

The first two terms are independent-sources sums; Slutsky's lemma and the CLT of Step 3 give

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) \xrightarrow{d} N\left(\mathbf{0}, \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}^{-1} (\mathbf{V}^{\text{data}} + \frac{1}{B} \mathbf{V}^{\text{sim}}) \mathbf{H}^{-1} \mathbf{M}^\top \mathbf{v}_{-1}\right),$$

where $\mathbf{V}^{\text{sim}}/B \rightarrow \mathbf{0}$ as $B \rightarrow \infty$. This is (S7). □ □

Corollary S1 (Variance estimation). *Under the conditions of Theorem S2, a consistent estimator of $\Sigma_{\text{A-SIMEX}}$ is obtained by replacing $\mathbf{H}(\lambda_j)$ with its sample analogue, and $\mathbf{V}^{\text{data}}, \mathbf{V}^{\text{sim}}$ with the empirical covariance matrices of the vectors $\{\mathbb{E}[\mathbf{m}_{ib} \mid Y_i, X_i, \boldsymbol{\varepsilon}_i]\}_i$ and $\{\mathbf{m}_{ib} - \mathbb{E}[\mathbf{m}_{ib} \mid Y_i, X_i, \boldsymbol{\varepsilon}_i]\}_{i,b}$ (the conditional expectation is estimated by the sample average over $b = 1, \dots, B$ for each i). Equivalently, the pairs-bootstrap over observations i (holding the simulated errors fixed and redrawing both data and simulated errors in each bootstrap world) is consistent for $\Sigma_{\text{A-SIMEX}}$.*

Proof. Each component converges by the same uniform LLN as in Lemma S2 (the influence scores have finite second moments uniformly in j, b ; the conditional-mean estimator averages B independent terms, giving error $O_p((nB)^{-1/2}) = o_p(n^{-1/2})$). The bootstrap claim follows from standard M-estimation bootstrap theory applied to the stacked influence representation of Step 2 of Theorem S2. Details are routine and omitted. $\square$

S2.0.7 Theorem 4: Asymptotic normality in the growing-degree regime

This section reconciles the growing-degree regime required for consistency (Assumption S4) with a central limit theorem centered at β_0 , closing the gap left by Remark S3 of the earlier draft. Throughout, the grid is equispaced on $[0, \bar{\lambda}]$ with $K = K(n)$ points and the extrapolation degree is $d = d(n)$. Map $[0, \bar{\lambda}]$ affinely to $[-1, 1]$ by $t(\lambda) = 2\lambda/\bar{\lambda} - 1$, so that the extrapolation point $\lambda = -1$ maps to

$$t_0 = t(-1) = -1 - \frac{2}{\bar{\lambda}} < -1, \quad \kappa := |t_0| + \sqrt{t_0^2 - 1} = 1 + \frac{2}{\bar{\lambda}} + \sqrt{\left(1 + \frac{2}{\bar{\lambda}}\right)^2 - 1} > 1.$$

The constant κ is the usual Bernstein-parameter of the point t_0 outside $[-1, 1]$: by the classical Chebyshev extremal property, any polynomial q of degree k satisfies $|q(t_0)| \leq \kappa^k \sup_{t \in [-1, 1]} |q(t)|$ (Rivlin, 1969, Ch. 3). Recall from Lemma S3 that $\beta(\cdot)$ is real-analytic on a neighbourhood of $[-1, \bar{\lambda}]$ and admits degree- k approximation error $O(\rho^{-k})$ on $[-1, \bar{\lambda}]$, where $\rho > 1$ depends on the analyticity strip of β ; the same holds for the influence functions introduced below.

Lemma S5 (Extrapolation operator bounds). *Let $K \geq c_0 d^2$ and let $\lambda_0 < \dots < \lambda_{K-1}$ be equispaced on $[0, \bar{\lambda}]$. Let $\{\varphi_k\}_{k=0}^d$ be the polynomials orthonormal with respect to the discrete inner product $\langle f, g \rangle_K = \frac{1}{K} \sum_j f(\lambda_j) g(\lambda_j)$, and let*

$$L_{K,d}(h) := \sum_{j=0}^{K-1} c_j h(\lambda_j), \quad c_j = \sum_{k=0}^d \varphi_k(t_0) \varphi_k(\lambda_j),$$

be the linear functional “least-squares degree- d fit evaluated at $\lambda = -1$ ”. Then:

- (i) $\sum_{k=0}^d \varphi_k(t_0)^2 \leq C d \kappa^{2d}$.
- (ii) $\sum_j |c_j| \leq C d^{1/2} K \kappa^d$ and $L_{K,d}(q) = q(-1)$ for every polynomial q of degree $\leq d$ (exactness on the model space).
- (iii) For every h real-analytic on a neighbourhood of $[-1, \bar{\lambda}]$ with analytic continuation bounded by M on the Bernstein ellipse E_ρ (in t -coordinates),

$$|L_{K,d}(h) - h(-1)| \leq C M d^{3/2} K \kappa^d \rho^{-d}.$$

Proof. (i) The discrete and continuous (uniform measure on $[0, \bar{\lambda}]$) inner products agree on polynomials of degree $\leq 2d$ up to relative error $O(d^2/K)$: for $p \geq 0$ of degree $\leq 2d$, $|\frac{1}{K} \sum_j p(\lambda_j) - \int p d\mu| \leq (Cd^2/K) \int p d\mu$, a classical composite-quadrature estimate combining the Euler–Maclaurin error on each subinterval with the Markov inequality $\|p''\|_\infty \leq Cd^4\|p\|_\infty \leq Cd^4(d+1)^2 \int p d\mu$ (Rivlin, 1969, Ch. 3; the last step is the L^1 – L^∞ Nikolskii inequality for degree- $2d$ polynomials). Consequently the Gram matrices of $\{1, t, \dots, t^d\}$ under the two inner products are equivalent with relative error $O(d^2/K) = O(1/c_0)$, so the discrete Christoffel function is within a constant factor of the continuous one:

$$\sum_{k=0}^d \varphi_k(t_0)^2 \leq C \sum_{k=0}^d (k + \tfrac{1}{2}) P_k(t_0)^2 \leq C d \kappa^{2d},$$

where P_k are the Legendre polynomials (orthonormal versions $\tilde{p}_k = \sqrt{k+1/2} P_k$), using $|P_k(t_0)| \leq \kappa^k$ from the Chebyshev extremal property and $\sum_{k \leq d} (k + \frac{1}{2}) \leq Cd^2$ against the dominant term κ^{2d} .

(ii) By the reproducing-kernel identity $c_j = \sum_k \varphi_k(t_0) \varphi_k(\lambda_j)$ and discrete orthonormality $\sum_j \varphi_k(\lambda_j) \varphi_{k'}(\lambda_j) = K \delta_{kk'}$,

$$\sum_j c_j^2 = \sum_{k,k'} \varphi_k(t_0) \varphi_{k'}(t_0) \sum_j \varphi_k(\lambda_j) \varphi_{k'}(\lambda_j) = K \sum_k \varphi_k(t_0)^2 \leq C d K \kappa^{2d},$$

whence $\sum_j |c_j| \leq \sqrt{K} (\sum_j c_j^2)^{1/2} \leq C d^{1/2} K \kappa^d$ by Cauchy–Schwarz. Exactness: $L_{K,d}$ is the evaluation at t_0 of the L^2 -orthogonal (discrete) projection onto $\mathcal{P}_d$, which reproduces every $q \in \mathcal{P}_d$; and evaluation is linear, so $L_{K,d}(q) = q(t_0) = q(-1)$ in λ -terms (since q as a polynomial in t maps to a polynomial of the same degree in λ and $t(-1) = t_0$).

(iii) Let q be the degree- d polynomial minimizing $\sup_{[-1, \bar{\lambda}]} |h - q| \leq CM \rho^{-d}$ (Chebyshev approximation of analytic functions; Lemma S3). Then

$$|L_{K,d}(h) - h(-1)| \leq |L_{K,d}(h - q)| + |q(-1) - h(-1)| \leq \left(\sum_j |c_j| \right) CM \rho^{-d} + CM \rho^{-d},$$

and (ii) gives the claim. $\square$

Define, for $\lambda \in [0, \bar{\lambda}]$, the data-influence function

$$\psi(\lambda) := \mathbb{E}[\mathbf{m}_{ib}(\lambda) \mid Y_i, X_i, \boldsymbol{\varepsilon}_i] = Y_i \mathbf{w}_i - \exp\left(\beta(\lambda)^\top \mathbf{w}_i + \frac{\lambda \sigma^2}{2} \beta(\lambda)^\top \boldsymbol{\Sigma}_0 \beta(\lambda)\right) (\mathbf{w}_i + \lambda \sigma^2 \boldsymbol{\Sigma}_0 \beta(\lambda)),$$

with $\mathbf{w}_i = X_i + \boldsymbol{\varepsilon}_i$ (this is the factorized conditional expectation of $\mathbf{m}_{ib}$ from the proof of Lemma S1; the exponential tilt is the λ -level augmentation with $U \sim N(0, \sigma^2 \boldsymbol{\Sigma}_0)$). By Lemma S1(ii), $\lambda \mapsto \beta(\lambda)$ is real-analytic on a neighbourhood of $[-1, \bar{\lambda}]$, hence so is $\psi(\lambda)$, and so is the covariance function

$$\boldsymbol{\Gamma}(\lambda, \lambda') := \text{Cov}(\psi_1(\lambda), \psi_1(\lambda')),$$

defined and real-analytic on a neighbourhood of $[-1, \bar{\lambda}]^2$. Write $\tilde{\boldsymbol{\Gamma}}$ for its analytic continuation and $\tilde{\boldsymbol{Z}} := \tilde{\boldsymbol{\Gamma}}(-1, -1)$; note $\boldsymbol{\Gamma}(\lambda, \lambda') \rightarrow \mathbf{I}_X(\boldsymbol{\beta}_0)$ as $\sigma^2 \rightarrow 0$ uniformly on $[-1, \bar{\lambda}]^2$ (by the same L^2 -collapse as in the proof of Theorem S4), so $\tilde{\boldsymbol{Z}} \rightarrow \mathbf{I}_X(\boldsymbol{\beta}_0)$.

Theorem S3 (Growing-degree CLT). *Grant Assumptions S1–S3, with $K = K(n)$ equispaced points and $B = B(n) \rightarrow \infty$. Suppose the selected degree satisfies $\hat{d}_n/d_n \xrightarrow{P} 1$ for a deterministic sequence*

$$d_n = \lceil c \log n \rceil, \quad K(n) \geq c_0 d_n^2,$$

and that the analyticity margin condition

$$\rho > \kappa \tag{S9}$$

holds with c chosen in the window $\frac{1}{2\log \rho} < c < \frac{1}{2\log \kappa}$ (nonempty exactly under (S9)). Then

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) \xrightarrow{d} N(\mathbf{0}, \Sigma_*), \quad \Sigma_* = \mathbf{I}_X(\beta_0)^{-1} \tilde{\mathbf{Z}} \mathbf{I}_X(\beta_0)^{-1},$$

with $\tilde{\mathbf{Z}}$ as defined above; the simulation component contributes $O(B^{-1})$ to Σ_* and vanishes as $B \rightarrow \infty$, and the extrapolation bias is $o(n^{-1/2})$, so the limit is centered at β_0 . In particular $\Sigma_* \rightarrow \Sigma_{\text{oracle}}$ as $\sigma^2 \rightarrow 0$, consistently with Theorem S4.

Remark S5. The window condition has a transparent interpretation: ρ measures how far the analytic continuation of the SIMEX curve extends (bias side), while κ measures how violently degree- d polynomial extrapolation to a point outside the grid amplifies noise (variance side); Theorem S3 is feasible precisely when the former dominates the latter. If (S9) fails, growing-degree extrapolation is variance-dominated and the fixed-degree CLT (Theorem S2) is the appropriate inference tool.

Proof of Theorem S3. Step 1: bias. The deterministic extrapolation error is $|L_{K,d_n}(\beta) - \beta(-1)| \leq C d_n^{3/2} K \kappa^{d_n} \rho^{-d_n}$ by Lemma S5(iii) applied to the (deterministic, analytic, M -bounded) curve β . In $\sqrt{n}$ units this is $\sqrt{n} d_n^{3/2} K \kappa^{d_n} \rho^{-d_n} = \text{polylog}(n) n^{1/2+c\log \kappa - c\log \rho} \rightarrow 0$ because $c(\log \rho - \log \kappa) > 0$. Hence centering at $\beta_0 = \beta(-1)$ is valid.

Step 2: uniform per-grid-point linearization. Exactly as in the proof of Theorem S2 (Step 1–2), with the maximal inequality entropy now carrying $\log(KB)$ and the quadratic Taylor remainder $n^{1/2}\delta_n^2 = O(\log(KBn)/n^{1/2}) = o(1)$ (using $K^2 \log n = o(n)$ and $B \leq n^{c'}$),

$$\sqrt{n}(\bar{\beta}(\lambda_j) - \beta(\lambda_j)) = \mathbf{H}(\lambda_j)^{-1} Z_n(\lambda_j) + o_p(1), \quad \text{uniformly in } j,$$

where $Z_n(\lambda) = n^{-1/2} \sum_i \bar{\mathbf{m}}_i(\lambda)$ and $\bar{\mathbf{m}}_i(\lambda) = B^{-1} \sum_b \mathbf{m}_{ib}(\lambda)$.

Step 3: empirical-process convergence with analytic limit paths. Decompose $\bar{\mathbf{m}}_i = \boldsymbol{\psi}_i + \mathbf{r}_i$ as in Theorem S2: $\mathbf{r}_i$ has conditional mean zero and variance $B^{-1} \mathbf{V}^{\text{sim}}$, so $\sup_\lambda \|n^{-1/2} \sum_i \mathbf{r}_i(\lambda)\| = O_p((nB)^{-1/2} \text{polylog}) = o_p(1)$ (the supremum over the grid is controlled by the same maximal inequality over the one-dimensional analytic family; the entropy of the class $\{\mathbf{m}_{ib}(\lambda) - \boldsymbol{\psi}(\lambda)\}$ is $O(\log(1/\delta))$). The class $\{\boldsymbol{\psi}(\lambda) : \lambda \in [0, \bar{\lambda}]\}$ is a one-dimensional analytic family with square-integrable envelope; its bracketing entropy is $O(\log(1/\delta))$, whence (i) the functional CLT

$$Z_n(\cdot) - n^{-1/2} \sum_i \mathbf{r}_i(\cdot) \Rightarrow Z(\cdot) \quad \text{in } \ell^\infty[0, \bar{\lambda}],$$

with Z a centered Gaussian process with covariance $\mathbf{\Gamma}$, and (ii) the sup-norm rate $\|Z_n - Z\|_\infty = O_p(\sqrt{\log n/n})$ (van der Vaart and Wellner, 1996, Thms. 2.14.1–2.14.2). The sample paths of Z are almost surely real-analytic on a neighbourhood of $[-1, \bar{\lambda}]$ with analytically continuable coordinates: this follows from the analyticity of the mean (zero) and covariance ($\mathbf{\Gamma}$) and the classical result that Gaussian processes with analytic covariance in a strip have a.s. analytic paths there (e.g. via the Cauchy–Kovalevskaya-type majorization of $\mathbb{E}\|\partial_\lambda^k Z\|^2 \leq C^k k!$, which ensures the Taylor series converges a.s.).

Step 4: reduction to a single linear functional. By Step 2,

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) = L_{K,d_n}(\mathbf{H}(\cdot)^{-1} Z_n(\cdot)) + o_p(1) + \underbrace{\sqrt{n} L_{K,d_n}(\beta - \beta(-1))}_{\rightarrow 0, \text{ Step 1}},$$

because $\hat{\beta}_{\text{A-SIMEX}} - \beta_0 = L_{K,d_n}(\bar{\beta} - \beta(-1))$ (exactness of L_{K,d_n} on polynomials, Lemma S5(ii)) and $\bar{\beta} - \beta(-1) = [\bar{\beta} - \beta] + [\beta - \beta(-1)]$ with the first bracket linearized in Step 2 and the second treated in Step 1.

Step 5: convergence of the functional. Write $\mathbf{g}_\omega(\lambda) = \mathbf{H}(\lambda)^{-1}Z(\lambda, \omega)$ for the (a.s. analytic) coordinate paths. On the event $E_n = \{\text{the analytic continuations of all } p \text{ coordinates of } \mathbf{g}_\omega \text{ are bounded by } M_n \text{ on } E_\rho\}$, where M_n is any sequence with $M_n \rightarrow \infty$ slowly (e.g. $M_n = \log \log n$), Lemma S5(iii) gives

$$\sup_{\omega \in E_n} |L_{K,d_n}(\mathbf{g}_\omega) - \mathbf{H}(-1)^{-1}\tilde{Z}(-1, \omega)| \leq CM_n d_n^{3/2} K \kappa^{d_n} \rho^{-d_n} \rightarrow 0,$$

by the same computation as in Step 1; and $\mathbb{P}(E_n) \rightarrow 1$ by the Borell–TIS inequality applied to the supremum of the analytic continuation of the Gaussian process over the ellipse (a.s. finite, with Gaussian concentration). The discretization error $L_{K,d_n}(\mathbf{H}^{-1}(Z_n - Z))$ satisfies

$$\|L_{K,d_n}(\mathbf{H}^{-1}(Z_n - Z))\| \leq \left(\sup_j \|\mathbf{H}(\lambda_j)^{-1}\| \right) \left(\sum_j |c_j| \right) \|Z_n - Z\|_\infty = O(d_n^{1/2} K \kappa^{d_n}) O_p\left(\sqrt{\frac{\log n}{n}}\right) \rightarrow 0,$$

because $d_n^{1/2} K \kappa^{d_n} \leq \text{poly}(d_n) n^{c \log \kappa} = o(n^{1/2}/\sqrt{\log n})$ under $2c \log \kappa < 1$. Finally, the linearization remainder of Step 2 propagated through L_{K,d_n} is $o_p(1)$ by the same operator bound.

Step 6: the limit law. Combining Steps 4–5,

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) = \mathbf{H}(-1)^{-1}\tilde{Z}(-1) + o_p(1),$$

and it remains to identify the law of the right-hand side. For each n the vector $\mathbf{H}(-1)^{-1}L_{K,d_n}(\mathbf{H}^{-1}Z_n)$ is a centered sample average of i.i.d. bounded- $(2 + \nu)$ -moment vectors divided by $\sqrt{n}$, with per-observation contributions bounded by $O(d_n^{1/2} K \kappa^{d_n})/\sqrt{n} = o(1)$ times a fixed sub-exponential variable; the Lindeberg condition therefore holds, and its variance converges by the double functional convergence

$$\text{Cov}\left(L_{K,d_n}(\mathbf{H}^{-1}Z_n)\right) = \sum_{j,k} c_j c_k \mathbf{H}_j^{-1} \Gamma(\lambda_j, \lambda_k) \mathbf{H}_k^{-1} \rightarrow \mathbf{H}(-1)^{-1} \tilde{\mathbf{Z}} \mathbf{H}(-1)^{-1},$$

applying Lemma S5(iii) twice to the real-analytic function $(\lambda, \lambda') \mapsto \mathbf{H}(\lambda)^{-1} \Gamma(\lambda, \lambda') \mathbf{H}(\lambda')^{-1}$ (exactness on polynomials is not needed here, only (iii), which applies to analytic functions of two variables by the same Chebyshev argument on the product ellipse). By the Cramér–Wold device the limit is $N(\mathbf{0}, \Sigma_*)$ with $\Sigma_* = \mathbf{I}_X^{-1} \tilde{\mathbf{Z}} \mathbf{I}_X^{-1}$, using $\mathbf{H}(-1) = \mathbf{H}(\beta_0, 0) = \mathbf{I}_X(\beta_0)$. The simulation component adds $O(B^{-1})$ to the variance, as in Theorem S2. $\square$ $\square$

S2.0.8 Theorem 3: Asymptotic relative efficiency

Let $\Sigma_{\text{oracle}} = \mathbf{I}_X(\beta_0)^{-1}$ with $\mathbf{I}_X(\beta_0) = \mathbb{E}[e^{X_i^\top \beta_0} X_i X_i^\top]$, and define the generalized asymptotic relative efficiency

$$\text{ARE}(\text{A-SIMEX vs. Oracle}) = \frac{\det(\Sigma_{\text{oracle}})}{\det(\Sigma_{\text{A-SIMEX}})}.$$

Three information matrices must be distinguished. (a) The oracle information $\mathbf{I}_X(\beta_0)$, for the model of (Y_i, X_i) . (b) The information $\mathbf{I}(t)$ of the *observed-data* model $(Y_i, \tilde{W}_t)$, $\tilde{W}_t = X_i + \mathbf{e}_t$ with $\mathbf{e}_t \sim N(\mathbf{0}, t\Sigma_0)$, $t \geq 0$ (so $t = \sigma^2$ in the application); this is the information that bounds what *any* estimator based on noisy covariates can achieve. (c) The information $\mathbb{E}[e^{\tilde{W}^\top \beta_0} \tilde{W} \tilde{W}^\top]$ of the *naive working model*, which is what the draft denoted $\mathbf{I}_W$ and used as Σ_{param} ; it is a third object, the curvature of a misspecified likelihood, and plays no role in the correct theory (see Remark S4).

Lemma S6 (Information monotonicity under additive noise). *Grant Assumptions S1 with Σ_0 fixed. Then:*

(i) *For $0 \leq t_1 \leq t_2$, $\mathbf{I}(t_2) \preceq \mathbf{I}(t_1)$ in the Loewner order; in particular $\mathbf{I}(t) \preceq \mathbf{I}(0) = \mathbf{I}_X(\beta_0)$ for all $t \geq 0$.*

(ii) *The inequality is strict for $t > 0$: $\mathbf{I}(t) \prec \mathbf{I}_X(\beta_0)$ provided X_i is not a.s. constant.*

Consequently $\Sigma_{\text{oracle}} = \mathbf{I}_X(\beta_0)^{-1} \preceq \mathbf{I}(t)^{-1}$ for all $t \geq 0$: the oracle covariance is the smallest, and by the Hájek convolution theorem $\mathbf{I}(t)^{-1}$ is the best asymptotic covariance achievable by any regular estimator based on $(Y_i, \widetilde{W}_t)$ alone.

Proof. For $t_2 \geq t_1$ write $\widetilde{W}_{t_2} = \widetilde{W}_{t_1} + \boldsymbol{\eta}$ with $\boldsymbol{\eta} \sim N(\mathbf{0}, (t_2 - t_1)\Sigma_0)$ independent of $(Y_i, \widetilde{W}_{t_1})$; the model of $(Y, \widetilde{W}_{t_2})$ is thus the image of the model of $(Y, \widetilde{W}_{t_1})$ under a Markov kernel that does not depend on β . Standard score transformation under parameter-independent smoothing: if $Z \sim P_\beta$ and $Z' = Z + \boldsymbol{\eta}$ with $\boldsymbol{\eta} \perp \beta$, then the score of Z' equals the conditional expectation of the score of Z given Z' ,

$$\mathbf{s}_{Z'}(\beta) = \mathbb{E}[\mathbf{s}_Z(\beta) \mid Z'],$$

justified by differentiating $p_{Z'}(z') = \int k(z' \leftarrow z) p_Z(z) dz$ under the integral: domination holds because the Gaussian kernel supplies exponential moments and the score $\mathbf{s}_Z(\beta) = [Y - e^{\beta^\top X}]X$ (for $Z = (Y, X)$) has uniformly bounded exponential moments. Then, by the law of total variance,

$$\mathbf{I}(t_2) = \text{Var}(\mathbb{E}[\mathbf{s}_Z \mid Z']) \preceq \text{Var}(\mathbf{s}_Z) = \mathbf{I}(t_1),$$

with equality iff $\mathbf{s}_Z$ is Z' -measurable. For $t_1 = 0$, $\mathbf{s}_Z = [Y - e^{\beta_0^\top X}]X$ with $Z = (Y, X)$; if equality held for some $t_2 > 0$, $\mathbf{s}_Z$ would be measurable with respect to $(Y, X + \mathbf{e}_{t_2})$, which is impossible unless X is a.s. constant (the conditional law of X given $(Y, X + \mathbf{e}_t)$ is non-degenerate for $t > 0$ on a set of positive probability when X is not a.s. constant, and $\mathbf{s}_Z$ genuinely depends on X). This proves (i)–(ii). The final display follows by inverting the Loewner order (for positive definite matrices, $\mathbf{0} \prec \mathbf{A} \preceq \mathbf{B}$ implies $\mathbf{B}^{-1} \preceq \mathbf{A}^{-1}$) and invoking the Hájek convolution theorem for the local limit of regular estimators in the model of $(Y, \widetilde{W}_t)$. $\square$

Theorem S4 (Efficiency relative to the oracle). *Grant the conditions of Theorem S2 with $B \rightarrow \infty$, and suppose $\Sigma_\epsilon = \sigma^2 \Sigma_0$ with Σ_0 fixed and $\sigma^2 \downarrow 0$ (with all other quantities, including the grid and the law of X_i , held fixed). Write $\Sigma_{\text{A-SIMEX}}(\sigma^2)$ for the asymptotic covariance of Theorem S2 at measurement-error scale σ^2 . Then:*

(i) **(First-order efficiency.)** $\Sigma_{\text{A-SIMEX}}(\sigma^2) \rightarrow \Sigma_{\text{oracle}}$ as $\sigma^2 \rightarrow 0$, for every fixed grid and degree: the limit is exact rather than merely $O(\sigma^2)$ -close.

(ii) **(Explicit second-order term.)** $\Sigma_{\text{A-SIMEX}}(\sigma^2) = \Sigma_{\text{oracle}} + \sigma^2 \dot{\mathbf{C}}_\beta + O(\sigma^4)$, where $\dot{\mathbf{C}}_\beta$ is the symmetric, explicitly computable matrix

$$\dot{\mathbf{C}}_\beta = \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}_0^{-1} \left(\dot{\mathbf{V}} - \dot{\mathbf{H}} \mathbf{H}_0^{-1} \mathbf{V}_0 - \mathbf{V}_0 \mathbf{H}_0^{-1} \dot{\mathbf{H}} \right) \mathbf{H}_0^{-1} \mathbf{M}^\top \mathbf{v}_{-1}, \quad (\text{S10})$$

with $\mathbf{H}_0 = \mathbf{I}_K \otimes \mathbf{I}_X(\beta_0)$, $\mathbf{V}_0 = (\mathbf{1}_K \mathbf{1}_K^\top) \otimes \mathbf{I}_X(\beta_0)$, and

$$\dot{\mathbf{H}}_j = \frac{\delta_j}{2} (\beta_0^\top \Sigma_0 \beta_0) \mathbf{I}_X + \delta_j \mathbb{E}[e^{\beta_0^\top X} (\tilde{\mathbf{b}}_{\Sigma_0}^\top X) X X^\top] + \Sigma_0 \beta_0 \mathbf{g}_0^\top + \mathbf{g}_0 \beta_0^\top \Sigma_0, \quad (\text{S11})$$

$$\dot{\mathbf{V}}_{jk} = \text{Cov}(\dot{\psi}_j, \psi^{(0)}) + \text{Cov}(\psi^{(0)}, \dot{\psi}_k), \quad \dot{\psi}_j = -e^{\beta_0^\top X} \left[(\delta_j \tilde{\mathbf{b}}_{\Sigma_0}^\top X + \frac{\lambda_j}{2} \beta_0^\top \Sigma_0 \beta_0) X + \lambda_j \Sigma_0 \beta_0 \right], \quad (\text{S12})$$

where $\delta_j = 1 + \lambda_j$, $\mathbf{g}_0 = \mathbb{E}[e^{\beta_0^\top X} X]$, and $\psi^{(0)} = YX - e^{\beta_0^\top X} X$ is the oracle influence function. Symmetry of $\dot{\mathbf{H}}_j$ and $\dot{\mathbf{V}}$ is immediate from (S11)–(S12), and (S10) is a congruence-like transform of symmetric matrices, hence $\dot{\mathbf{C}}_\beta$ is symmetric. In general, however, $\dot{\mathbf{C}}_\beta$ is indefinite: neither $\dot{\mathbf{C}}_\beta \succeq \mathbf{0}$ nor $\dot{\mathbf{C}}_\beta \preceq \mathbf{0}$ holds universally, and the sign of $\text{tr}(\Sigma_{\text{oracle}}^{-1} \dot{\mathbf{C}}_\beta)$ depends on the design, the grid, and the degree (see Remark S6).

(iii) (**Determinant expansion and correct efficiency statement.**)

$$\text{ARE} = 1 + \sigma^2 \kappa_\beta + O(\sigma^4), \quad \kappa_\beta := -\text{tr}(\Sigma_{\text{oracle}}^{-1} \dot{\mathbf{C}}_\beta),$$

with no universal sign on κ_β . The universal statements are: (a) $\text{ARE} \rightarrow 1$ as $\sigma^2 \rightarrow 0$ (part (i)); (b) by Lemma S6, $\Sigma_{\text{oracle}} \preceq \mathbf{I}(\sigma^2)^{-1}$, so no regular estimator based on the noisy covariates alone can improve on the oracle to first order; and (c) A-SIMEX attains this frontier to first order, its excess over Σ_{oracle} being second-order ($O(\sigma^2)$) with a bias–variance trade-off reflected in the sign of κ_β .

Remark S6 (Why the draft’s positive semidefiniteness claim is false). The draft asserted $\dot{\mathbf{C}}_\beta \succeq \mathbf{0}$, hence $\text{ARE} \leq 1 + O(\sigma^4)$. This is not correct, for a structural reason: measurement error attenuates the signal but can also *de-noise* the estimating directions. In the homoskedastic linear model $Y = X\beta_0 + e$, $W = X + \epsilon$, the naive slope has asymptotic variance $\sigma_e^2/\mathbb{E}[W^2] < \sigma_e^2/\mathbb{E}[X^2]$, the oracle variance: the pseudo-estimators’ variances *decrease* in σ^2 , and the resulting negative contribution enters $\dot{\mathbf{V}}$ through (S12). Since A-SIMEX is a fixed linear combination ($\sum_j c_j = 1$) of these pseudo-estimators, its second-order variance term inherits this sign. The Poisson log-link exhibits the same phenomenon for suitable designs. What survives universally is part (iii): the oracle marks the first-order frontier (Lemma S6), A-SIMEX reaches it, and the second-order gap splits into the extrapolation bias $\sigma^2(\sum_j c_j \delta_j) \tilde{\mathbf{b}}_{\Sigma_0}$ (deterministic, $O(\sigma^2)$, vanishing for grids with $\sum_j c_j \delta_j = 0$) plus the symmetric variance correction (S10) of indefinite sign.

Corollary S2 (Explicit scalar constants). For $p = 1$, the first-order naive bias is, by (S2),

$$\tilde{b} = -\frac{\frac{1}{2}\beta_0^2 \mu_1(\beta_0) + \beta_0 \mu_0(\beta_0)}{\mu_2(\beta_0)} \cdot \sigma^2\text{-scale}, \quad \mu_r(\beta) = \mathbb{E}[e^{\beta X} X^r],$$

so for $X \sim N(0, 1)$, $\tilde{b} = -\sigma^2 \beta_0(1 + \beta_0^2/2)/(1 + \beta_0^2)$ (attenuation). The second-order constant is

$$\kappa_\beta = -\frac{1}{\mu_2(\beta_0)} \sum_{j,k} c_j c_k \left[\dot{\Gamma}_{jk} - \dot{H}_j \Gamma_{jk}^{(0)} \cdot \frac{1}{\mu_2(\beta_0)} - \Gamma_{jk}^{(0)} \dot{H}_k \cdot \frac{1}{\mu_2(\beta_0)} \right],$$

with $\Gamma_{jk}^{(0)} = \text{Var}(\psi^{(0)}) = \mu_2(\beta_0)$, $\dot{H}_j$ from (S11) (scalar), and $\dot{\Gamma}_{jk}$ from (S12) (scalar); every entry is an explicit functional of $\{\mu_r(\beta_0)\}_{r \leq 4}$ and the grid weights $\{c_j\}$.

Proof of Theorem S4. Step 1: the $\sigma^2 \rightarrow 0$ collapse of the influence structure. As $\sigma^2 \downarrow 0$, for every grid point λ_j , $\beta(\lambda_j) \rightarrow \beta(0) = \beta_0$ (Lemma S1(ii)–(iii) with $\delta_j = 1 + \lambda_j \rightarrow 1$, and $\beta(\delta) \rightarrow \beta(0)$ by continuity; note $\beta(0) = \beta_0$ corresponds to the oracle). The influence score at λ_j ,

$$\mathbf{m}_{ib}(\lambda_j) = [Y_i - e^{\beta(\lambda_j)^\top X_i} W_i^{(b)}] W_i^{(b)}, \quad W_i^{(b)} = X_i + \epsilon_i + \sqrt{\lambda_j} U_{ib},$$

converges in L^2 to $\mathbf{m}_i(0) = [Y_i - e^{\beta_0^\top X_i}] X_i$, because $W_i^{(b)} \rightarrow X_i$ in L^4 (Gaussian tails), $e^{\beta(\lambda_j)^\top X_i} W_i^{(b)} \rightarrow e^{\beta_0^\top X_i}$ in L^2 (locally uniform convergence plus dominated convergence with the sub-exponential dominant $e^{C\|W_i^{(b)}\|}$, whose moments are bounded uniformly in $\sigma^2 \leq \bar{\sigma}^2$), and the products converge

by Cauchy–Schwarz. Hence the stacked covariance converges to $\mathbf{V}_0 = (\mathbf{1}_K \mathbf{1}_K^\top) \otimes \mathbf{I}_X(\boldsymbol{\beta}_0)$ (perfect cross-grid correlation: all K influence vectors collapse to the same oracle influence) and $\mathbf{H} \rightarrow \mathbf{I}_K \otimes \mathbf{I}_X(\boldsymbol{\beta}_0) = \mathbf{H}_0$.

Step 2: the extrapolation functional becomes the identity. The extrapolated estimator is $\hat{\beta}_{\text{A-SIMEX}} = \sum_j c_j \bar{\beta}(\lambda_j)$ with $\sum_j c_j = 1$. Substituting the limit covariance,

$$\boldsymbol{\Sigma}_{\text{A-SIMEX}} \rightarrow \sum_{j,k} c_j c_k \cdot \mathbf{I}_X(\boldsymbol{\beta}_0)^{-1} \mathbf{I}_X(\boldsymbol{\beta}_0) \mathbf{I}_X(\boldsymbol{\beta}_0)^{-1} = \left(\sum_j c_j \right)^2 \mathbf{I}_X(\boldsymbol{\beta}_0)^{-1} = \boldsymbol{\Sigma}_{\text{oracle}},$$

since $\sum_j c_j = 1$ (the constant function lies in the span of the basis). Equivalently, in matrix form: $\mathbf{H}_0^{-1} \mathbf{V}_0 \mathbf{H}_0^{-1} = (\mathbf{1}\mathbf{1}^\top) \otimes \mathbf{I}_X^{-1}$ and $\mathbf{v}_{-1}^\top \mathbf{M}((\mathbf{1}\mathbf{1}^\top) \otimes \mathbf{I}_X^{-1}) \mathbf{M}^\top \mathbf{v}_{-1} = (\sum_j c_j)^2 \mathbf{I}_X^{-1} = \mathbf{I}_X^{-1}$. This proves (i).

Step 3: first-order expansion of the ingredients. All σ^2 -dependence enters through (a) the pseudo-true curve $\beta(\lambda_j; \sigma^2) = \boldsymbol{\beta}_0 + \delta_j \sigma^2 \tilde{\mathbf{b}}_{\boldsymbol{\Sigma}_0} + O(\sigma^4)$ (Lemma S1(iii)), with

$$\tilde{\mathbf{b}}_{\boldsymbol{\Sigma}_0} = -\mathbf{I}_X(\boldsymbol{\beta}_0)^{-1} \left[\frac{1}{2} (\boldsymbol{\beta}_0^\top \boldsymbol{\Sigma}_0 \boldsymbol{\beta}_0) \mathbb{E}[e^{\boldsymbol{\beta}_0^\top X} X] + \boldsymbol{\Sigma}_0 \boldsymbol{\beta}_0 \mathbb{E}[e^{\boldsymbol{\beta}_0^\top X}] \right],$$

(b) the Gaussian tilt factor $e^{(\lambda_j \sigma^2 / 2) \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_0 \boldsymbol{\beta}}$ in the conditional mean of the influence scores, and (c) the tilt of the design moment $\mathbb{E}[e^{\boldsymbol{\beta}^\top (X + \boldsymbol{\varepsilon}_\delta)} (X + \boldsymbol{\varepsilon}_\delta)(X + \boldsymbol{\varepsilon}_\delta)^\top]$. For (c), the factorization

$$\mathbf{H}_j(\sigma^2) = e^{\frac{\delta_j \sigma^2}{2} \boldsymbol{\beta}_j^\top \boldsymbol{\Sigma}_0 \boldsymbol{\beta}_j} \left[\mathbf{H}(\boldsymbol{\beta}_j, 0) + \delta_j \sigma^2 (\boldsymbol{\Sigma}_0 \boldsymbol{\beta}_j \mathbf{g}_j^\top + \mathbf{g}_j \boldsymbol{\beta}_j^\top \boldsymbol{\Sigma}_0) \right],$$

(where $\mathbf{H}(\boldsymbol{\beta}, 0) = \mathbb{E}[e^{\boldsymbol{\beta}^\top X} X X^\top]$, $\mathbf{g}_j = \mathbb{E}[e^{\boldsymbol{\beta}_j^\top X} X]$, and we used $\mathbb{E}[e^{\boldsymbol{\beta}^\top \boldsymbol{\varepsilon}_\delta} \boldsymbol{\varepsilon}_\delta] = \delta \sigma^2 \boldsymbol{\Sigma}_0 \boldsymbol{\beta} e^{(\delta \sigma^2 / 2) \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_0 \boldsymbol{\beta}}$) gives, after differentiating in σ^2 at 0 and using $\boldsymbol{\beta}_j(\sigma^2) = \boldsymbol{\beta}_0 + \delta_j \sigma^2 \tilde{\mathbf{b}}_{\boldsymbol{\Sigma}_0}$, exactly (S11). For (a)–(b), the conditional mean of the influence score,

$$\boldsymbol{\psi}_j(\sigma^2) = Y_i \mathbf{w}_i - \exp\left(\boldsymbol{\beta}_j(\sigma^2)^\top \mathbf{w}_i + \frac{\lambda_j \sigma^2}{2} \boldsymbol{\beta}_j(\sigma^2)^\top \boldsymbol{\Sigma}_0 \boldsymbol{\beta}_j(\sigma^2)\right) (\mathbf{w}_i + \lambda_j \sigma^2 \boldsymbol{\Sigma}_0 \boldsymbol{\beta}_j(\sigma^2)),$$

is real-analytic in σ^2 with $\boldsymbol{\psi}_j(0) = Y_i \mathbf{w}_i - e^{\boldsymbol{\beta}_0^\top \mathbf{w}_i} \mathbf{w}_i =: \boldsymbol{\psi}^{(0)}$ (independent of j), and differentiating at $\sigma^2 = 0$ yields (S12) for $\dot{\mathbf{V}}_{jk} = \text{Cov}(\dot{\boldsymbol{\psi}}_j, \boldsymbol{\psi}^{(0)}) + \text{Cov}(\boldsymbol{\psi}^{(0)}, \dot{\boldsymbol{\psi}}_k) = \frac{d}{d\sigma^2} \text{Cov}(\boldsymbol{\psi}_j, \boldsymbol{\psi}_k) \big|_{\sigma^2=0}$ (the derivative passes under Cov by dominated convergence, all moments being uniformly bounded on $\sigma^2 \leq \bar{\sigma}^2$). Both $\dot{\mathbf{H}}$ and $\dot{\mathbf{V}}$ are symmetric: for $\dot{\mathbf{V}}$ this is the general fact $\frac{d}{dt} \text{Cov}(\boldsymbol{\psi}_t) = \text{Cov}(\dot{\boldsymbol{\psi}}_t, \boldsymbol{\psi}_t) + \text{Cov}(\boldsymbol{\psi}_t, \dot{\boldsymbol{\psi}}_t)$.

Step 4: propagation through the sandwich. The asymptotic covariance of Theorem S2 is the smooth matrix-valued map $\mathbf{S}(\sigma^2) = \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}(\sigma^2)^{-1} \mathbf{V}(\sigma^2) \mathbf{H}(\sigma^2)^{-1} \mathbf{M}^\top \mathbf{v}_{-1}$. Differentiating $\mathbf{H}^{-1} \mathbf{V} \mathbf{H}^{-1}$ at a point where $\mathbf{H}_0^{-1} \mathbf{V}_0 \mathbf{H}_0^{-1} = (\mathbf{1}\mathbf{1}^\top) \otimes \mathbf{I}_X^{-1}$ (Step 2) gives

$$\frac{d}{d\sigma^2} \bigg|_0 \mathbf{H}^{-1} \mathbf{V} \mathbf{H}^{-1} = \mathbf{H}_0^{-1} \left(\dot{\mathbf{V}} - \dot{\mathbf{H}} \mathbf{H}_0^{-1} \mathbf{V}_0 - \mathbf{V}_0 \mathbf{H}_0^{-1} \dot{\mathbf{H}} \right) \mathbf{H}_0^{-1},$$

whence (S10). The deterministic extrapolation bias does not enter $\boldsymbol{\Sigma}_{\text{A-SIMEX}}$ (a covariance): it equals $\sigma^2 (\sum_j c_j \delta_j) \tilde{\mathbf{b}}_{\boldsymbol{\Sigma}_0} + O(\sigma^4)$ and affects the risk, not the variance; we account for it explicitly in Remark S6 and part (iii).

Step 5: determinant expansion. With $\boldsymbol{\Sigma}_{\text{A-SIMEX}} = \boldsymbol{\Sigma}_{\text{oracle}} + \sigma^2 \dot{\mathbf{C}}_\beta + O(\sigma^4)$ and both leading matrices positive definite/symmetric,

$$\det(\boldsymbol{\Sigma}_{\text{A-SIMEX}}) = \det(\boldsymbol{\Sigma}_{\text{oracle}}) \left(1 + \sigma^2 \text{tr}(\boldsymbol{\Sigma}_{\text{oracle}}^{-1} \dot{\mathbf{C}}_\beta) + O(\sigma^4) \right),$$

which is part (iii). No sign claim is made on κ_β , per Remark S6. This proves (ii)–(iii).

Proof of Corollary S2. The scalar pseudo-score condition is (Lemma S1(i) and the tilt identities in its proof)

$$\mu_1(\beta_0) = e^{\frac{\delta}{2}\beta^2\sigma^2}(\mu_1(\beta) + \delta\sigma^2\beta\mu_0(\beta)), \quad \mu_r(\beta) = \mathbb{E}[e^{\beta X}X^r],$$

so $\beta(\delta) = \beta_0 + \delta\sigma^2\tilde{b} + O(\sigma^4)$ with $\tilde{b} = -[\frac{1}{2}\beta_0^2\mu_1(\beta_0) + \beta_0\mu_0(\beta_0)]/\mu_2(\beta_0) < 0$ for $\beta_0 > 0$; for $X \sim N(0, 1)$, $\mu_0(\beta) = e^{\beta^2/2}$, $\mu_1(\beta) = \beta e^{\beta^2/2}$, $\mu_2(\beta) = (1 + \beta^2)e^{\beta^2/2}$, giving $\tilde{b} = -\sigma^2\beta_0(1 + \beta_0^2/2)/(1 + \beta_0^2)$, matching Lemma S1(iii). The scalar specialization of (S10) is displayed in the corollary; each ingredient ($\dot{H}_j$, $\dot{\Gamma}_{jk}$) is obtained from (S11)–(S12) by replacing expectations with $\mu_r(\beta_0)$ -combinations (the fourth moment μ_4 enters through $\text{Cov}(\psi_j, \psi^{(0)})$, which involves $\mathbb{E}[e^{2\beta_0 X}X^r]$ -type terms reducible to $\{\mu_r(\beta_0)\}_{r \leq 4}$ by completing the square). $\square$ $\square$

S3 Implementation Details

S4 Implementation Details

Package Structure

adaptsimex/	
R/main.R	main <code>adaptive_simex()</code> driver
R/cv_select.R	cross-validation for $(\lambda_{\max}, \text{degree})$ (prefix property, 1-SE rule)
R/variance.R	analytic sandwich (equation 3.4 of the main text) and pairs-bootstrap
R/methods.R	S3 methods: print, summary, confint, plot
src/simex_batch.cpp	optional Rcpp batching of the replicate-level IRLS
vignettes/	quick start; theory; NHANES case study
tests/testthat/	unit tests (attenuation direction, sandwich vs. bootstrap)

Main Function and Usage

```
adaptive_simex(formula, data, me_vars, sigma_error,
lambda_max_seq = NULL, poly_degree_seq = 1:3,
n_simex = 100, n_cv_folds = 5,
se_method = c("sandwich", "bootstrap"), n_boot = NULL,
parallel = TRUE, n_cores = NULL, verbose = TRUE)
Returns an object of class c("adaptive_simex", "glm") with slots coefficients, vcov, se,
selected_lambda_max, selected_poly_degree, cv_results, and call.
```

Here `me_vars` names the error-contaminated columns and `sigma_error` their standard deviations; `se_method = "sandwich"` uses the analytic sandwich, and `"bootstrap"` the pairs-bootstrap recommended at small n (main text, Section 5.7). A typical call:

```
fit <- adaptive_simex(Y ~ W1 + W2 + X3, data = df,
me_vars = c("W1", "W2"), sigma_error = c(0.3, 0.3),
n_simex = 100, se_method = "sandwich", parallel = TRUE)
summary(fit); confint(fit); plot(fit)
```

Computational Complexity and Parallelization

Three implementation ideas reduce the cost far below naive estimates. First, all B SIMEX replicates at a grid point are fitted by one *batched* IRLS sweep: the score and Hessian accumulations are tensor contractions over a (B, n, p) array, so the B fits share the linear-algebra machinery. Second, the CV procedure exploits the *prefix property* of the SIMEX grid: one union-grid curve per fold (with noise

shared across λ within a replicate, per Algorithm 1) serves all candidate $\lambda_{\max}$, reducing the number of fitted pseudo-samples per fold from $|\mathcal{K}_\lambda| \cdot |\Lambda| B_{\text{cv}}$ to $|\Lambda| B_{\text{cv}}$. Third, fits along the curve are warm started from the previous grid point. Measured single-core times for the full A-SIMEX fit (CV plus final curve plus sandwich) are 0.10, 0.28, 0.95, and 3.3 seconds at $(n=200, p) \in \{5, 15, 30, 50\}$ with $B = 100$, $B_{\text{cv}} = 25$; standard SIMEX is about twice as fast, and the corrected-score estimator is one to two orders of magnitude cheaper but numerically fragile (main text, Section 5.3). The asymptotic complexity per replication remains $O(B |\Lambda| np^2)$ for the final curve plus $O(5 |\Lambda| B_{\text{cv}} np^2)$ for the CV, but the constants are small enough that parallelization over Monte Carlo repetitions, CV folds, or grid points, all of them embarrassingly parallel and all supported by the package through the `future/furrr` backends, is a matter of convenience rather than necessity on modern hardware.

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